

# Sean's Number Simulation Script

This reference file is for rebuilding only the Sean's Number narrative simulation and post-outcome analysis/debrief. Keep the personality test, result page, and challenge-selection flow unchanged.

## Design goal

This scenario should feel like a short narrative classroom-management game: playful enough for teachers to explore, but specific enough that the debrief teaches them how their choices affected student thinking, mathematical clarity, and classroom discussion norms.

This version makes the answer choices more consequential, fixes the outcome-path logic, and makes the result page immediately show whether the player reached a failure, partial success, strong partial success, or success case.

## 1. Scenario basis

The simulation is adapted from the Sean/Ofala classroom transcript. Sean argues that 6 can be both odd and even because it can be made from three 2s and two 3s. Other students disagree. The teacher first returns to the class definition of even numbers. Mei then tests Sean's reasoning with 10. Ofala later proposes a stronger odd-number rule: after circling pairs of 2, odd numbers have one left over and even numbers do not.

### Core teaching problem

- Sean's claim sounds wrong, but it contains a real mathematical idea: he is noticing structure in how 6 can be composed.
- The teacher has to decide whether to correct the answer quickly or treat the misconception as a resource for class reasoning.
- For the game version, the weak paths should show what happens when the teacher closes the idea, lets debate run without structure, or gives the final answer too quickly.

### Main conceptual tension

- Sean is treating the number of groups as if it determines whether the whole number is odd or even.
- The class needs to distinguish between how objects are grouped and whether pairing by 2 leaves one leftover.
- The strongest resolution comes when students compare multiple representations: circles, groups, straight-line marks, pairings, and leftovers.

Best-practice arc: Student claim → peer disagreement → definition check → counterexample test → stronger odd/even rule → class tests the rule again.

## 2. Simulation UI / interaction specification

### Overall style

- Game type: short narrative story game, similar to a Japanese scout / visual-novel style classroom scenario.
- Setting: animated or softly rendered elementary classroom background. No real humans and no realistic human faces should appear in the background.
- Mood: warm, curious, slightly dramatic when disagreement appears, but never chaotic or punitive.
- Camera: mostly static classroom background for dialogue scenes. Switch to full-screen blackboard when math must be shown.

### Character and dialogue representation

- Each recurring speaker must have a consistent colored name tag. Use subtle, readable colors rather than neon colors.

- Narration appears as plain background-setting text with no quotation marks and no “Narrator:” label.
- Spoken lines must use quotation marks. Example: Sean: “I think 6 can be odd and even.”
- When a student or teacher is speaking, show only the name tag and text box. Do not generate human character portraits unless using abstract icons or simple colored avatars.

Suggested speaker tags

| Speaker              | Suggested tag color | Reason                                                                 |
|----------------------|---------------------|------------------------------------------------------------------------|
| Teacher              | Deep blue           | Stable facilitation voice; frames, redirects, and consolidates ideas.  |
| Sean                 | Orange              | Original student thinker; introduces the productive misconception.     |
| Cassandra            | Rose / red          | Challenge voice; makes disagreement visible.                           |
| Keith                | Purple              | Peer reasoning voice; pushes back without fully resolving the concept. |
| Mei                  | Teal                | Counterexample voice; tests Sean’s logic with another number.          |
| Ofala                | Green               | Definition-building voice; offers the pairing-and-leftover rule.       |
| Background narration | Neutral gray        | Sets classroom context without looking like a speaker.                 |

Full-screen blackboard mode

- Keep the full-screen blackboard background that is already working.
- The blackboard should become the main visual stage, not a small board floating behind text.
- Keep generous top padding so text and math visuals are not clipped at the top.
- Dialogue should sit in a separate lower area and must not cover circles, lines, labels, pairing marks, or leftover objects.
- Use simple chalk-style visuals: circles, tally-like lines, brackets, pairing loops, arrows, and short labels.

Required board visuals

- Sean’s grouped circles for  $6 = 2 + 2 + 2$ .
- Sean’s comparison with  $6 = 3 + 3$ .
- The teacher’s definition check: 6 can be split fairly into 3 and 3.
- Mei’s counterexample:  $10 =$  five groups of 2, showing why Sean’s logic overgeneralizes.
- Ofala’s straight-line visuals for odd numbers like 5 and 9.
- A final comparison: odd = pairs of 2 plus 1 leftover; even = pairs of 2 with 0 leftover.

3. Revised narrative flow and screen logic

| Stage                  | Player-facing story beat                                                                                                                           | Screen / visual behavior                                                                                                                                                                 |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Opening                | A class is discussing odd and even numbers after a previous meeting with another class. Sean raises his hand because he has been thinking about 6. | Normal classroom background. No blackboard yet unless a number line is subtly visible in the room.                                                                                       |
| Sean’s claim           | Sean says “6 can be odd and even” because it can be made from three 2s and two 3s.                                                                 | Switch to full-screen blackboard. Show 6 circles grouped as $2 + 2 + 2$ , then show $3 + 3$ .                                                                                            |
| Decision 1 consequence | The teacher either shuts the idea down, invites Sean to explain, or anchors his idea in a definition.                                              | If option A, stop the simulation immediately and show the failure result page. Do not continue to Choice 2 or Choice 3. If B or C, show Sean’s grouping so his thinking becomes visible. |
| Peer disagreement      | Cassandra and Keith disagree. The room shifts from sharing into a real mathematical disagreement.                                                  | Return to classroom dialogue. Use tension through pacing and dialogue, not visual chaos.                                                                                                 |

| Stage                  | Player-facing story beat                                                                                          | Screen / visual behavior                                                                               |
|------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Decision 2 consequence | The teacher either lets peers pile on, reviews the definition only, or tests Sean's logic with 10.                | For option C, full-screen blackboard shows 10 as five groups of 2.                                     |
| Definition-building    | Ofala proposes the stronger rule: after circling the twos, odd numbers have one left over.                        | Full-screen blackboard: show 5 and 9 with pairs plus one leftover; contrast with 6 having no leftover. |
| Decision 3 consequence | The teacher either gives the answer, summarizes the rule, or asks the class to test the rule with another number. | Only option C should lead to the full success ending because students apply the rule themselves.       |

## 4. Revised choice design

Each decision should include one weak/problematic move, one partial/incomplete move, and one strong concept-building move. The options should not all sound equally good.

### Choice 1: First response to Sean

| Option                        | Teacher move and effect                                                                                                                                              |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A - weak / problematic        | Close it down. "No, 6 is even. We already learned this. Let's move on."<br>Effect: protects accuracy but hides Sean's reasoning.                                     |
| B - partial / incomplete      | Invite Sean to explain. "Show us what you mean by three 2s and two 3s."<br>Effect: opens thinking, but still needs structure.                                        |
| C - strong / concept-building | Anchor and visualize. "Let's make your idea visible, then compare it with our definition of even."<br>Effect: combines student thinking with mathematical structure. |

### Choice 2: Guide the disagreement

| Option                        | Teacher move and effect                                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A - weak / problematic        | Let peers pile on. "Who thinks Sean is wrong? Tell him why."<br>Effect: creates debate, but weak mathematical anchor.                                                 |
| B - partial / incomplete      | Return only to definition. "Let's remember: even means it can split into two equal groups."<br>Effect: stabilizes the talk, but may not reveal why Sean's rule fails. |
| C - strong / concept-building | Test Sean's rule. "If group count decides oddness, what happens with 10 as five groups of 2?"<br>Effect: turns the misconception into a testable hypothesis.          |

### Choice 3: Close the discussion

| Option                        | Teacher move and effect                                                                                                                                            |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A - weak / problematic        | Teacher gives answer. "So the answer is 6 is even. That's the final rule."<br>Effect: ends cleanly but teacher-owned.                                              |
| B - partial / incomplete      | Teacher summarizes. "Sean noticed groups, but odd/even depends on whether there is a leftover after pairing by 2."<br>Effect: clarifies, but mostly teacher-owned. |
| C - strong / concept-building | Class tests the rule. "Choose another number. Predict the leftover, then test it with pairs."<br>Effect: students apply and own the rule.                          |

## 5. Exact outcome logic for implementation

Use this priority order in the code. This avoids the bug where  $B \rightarrow A \rightarrow A$  accidentally becomes Outcome 1. It also makes clear that Choice 1A is an immediate failure branch, not a path that continues through the remaining choices.

### Recommended logic:

- If Choice 1 = A, stop the simulation immediately and show Outcome 1: Premature Closure. Do not show Choice 2 or Choice 3.

- Else if Choice 2 = A, show Outcome 2: Messy Debate Without Anchor.
- Else if Choice 2 = B, show Outcome 3: Definition Recovery.
- Else if Choice 2 = C and Choice 3 = A or B, show Outcome 4: Productive Contradiction.
- Else if Choice 2 = C and Choice 3 = C, show Outcome 5: Conceptual Breakthrough.

The outcome should not be calculated by simply counting “good” choices. Use the branch logic above because each decision has a different instructional role.

| Path                 | Outcome                                | Performance level      | Why                                                                                                      |
|----------------------|----------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------|
| A - stop immediately | Outcome 1: Premature Closure           | Failure                | The first move shuts down Sean's reasoning before it becomes visible, so the simulation should end here. |
| B/C - A - any        | Outcome 2: Messy Debate Without Anchor | Failure                | The teacher opens talk, but lets the disagreement become peer rejection without an anchor.               |
| B/C - B - any        | Outcome 3: Definition Recovery         | Partial Success        | The teacher restores the definition, but does not fully test Sean's hidden rule.                         |
| B/C - C - A/B        | Outcome 4: Productive Contradiction    | Strong Partial Success | The teacher tests the misconception, but does not fully make the closing student-owned.                  |
| B/C - C - C          | Outcome 5: Conceptual Breakthrough     | Success                | The teacher helps students test, build, and apply the stronger rule.                                     |

## 6. Branch consequences inside the story

### Choice 1 branch consequences

| Choice 1                   | Immediate scene consequence                                                                                                                                                                                                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A - Close it down          | The teacher corrects Sean and keeps the lesson moving. Sean stops explaining. Some students nod, but the class never sees the hidden rule behind his mistake. The simulation should stop here and immediately show Outcome 1. |
| B - Invite Sean to explain | Sean draws three groups of 2. The class can see his thinking, but the discussion still needs a mathematical anchor.                                                                                                           |
| C - Anchor and visualize   | Sean's idea is placed on the board and immediately connected to the class definition. The scene keeps both student thinking and mathematical accuracy in view.                                                                |

### Peer disagreement beat

Cassandra: “I disagree. Six cannot be odd. It goes even, odd, even, odd, even, odd.”

Keith: “Just because two odd numbers can make 6 does not mean 6 has to be odd.”

This is where the game should make the teacher choice matter. If the player lets peer disagreement run without structure, the room becomes lively but less precise. If the player anchors or tests the idea, the room becomes more mathematically productive.

### Choice 2 branch consequences

| Choice 2                      | Immediate scene consequence                                                                                                                                    |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A - Let peers pile on         | Students respond with “No,” “That’s wrong,” and “Prove it.” Sean defends himself but becomes less clear. The class energy rises while the concept stays loose. |
| B - Return only to definition | The class reviews that even numbers can split into two equal groups. This helps, but Sean still argues that 6 can also be odd because it has three 2s.         |
| C - Test Sean's rule          | Mei introduces 10 as five groups of 2. The blackboard shows that Sean's logic would make 10 both odd and even too, creating a visible contradiction.           |

### Choice 3 branch consequences

| Choice 3                       | Immediate scene consequence                                                                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| A - Teacher gives final answer | The teacher ends the scene with the correct answer. The discussion stops, but student reasoning remains mostly teacher-controlled.                    |
| B - Teacher summarizes         | The teacher explains the distinction between group count and leftovers. The math becomes clearer, but students do not fully test the rule themselves. |
| C - Class tests the rule       | Ofala’s leftover rule becomes a shared method. Another student tests a new number, predicts the leftover, and checks it on the board.                 |

7. Result-page display rules

The result page should show the performance level first, then the outcome title. This makes the outcome immediately legible before the player reads the full explanation.

| Performance level      | Color cue       | Outcome title                                    | Meaning                                                                               |
|------------------------|-----------------|--------------------------------------------------|---------------------------------------------------------------------------------------|
| Failure                | #B3332E / red   | Premature Closure or Messy Debate Without Anchor | The player’s moves blocked or weakened concept-building.                              |
| Partial Success        | #B86B00 / amber | Definition Recovery                              | The player stabilized the math but did not fully resolve the hidden misconception.    |
| Strong Partial Success | #2F5D8C / blue  | Productive Contradiction                         | The player tested the misconception but did not fully transfer ownership to students. |
| Success                | #2E7D5B / green | Conceptual Breakthrough                          | The player helped students build and test a transferable rule.                        |

Recommended top-card format

- Line 1: Performance level, large and color-coded. Example: **Success**.
- Line 2: Specific outcome title. Example: Conceptual Breakthrough.
- Line 3: One-sentence summary. Example: “Students used Sean’s mistake to build a stronger rule about pairs and leftovers.”

8. Five outcomes

|                |                                                                                      |
|----------------|--------------------------------------------------------------------------------------|
| <b>Failure</b> | Outcome 1 - Premature Closure<br>Correct answer protected, student reasoning erased. |
|----------------|--------------------------------------------------------------------------------------|

- Typical path: Choice 1 = A, then stop immediately
- Key point: **The hidden misconception stays invisible.**
- Improve next: Before correcting, ask Sean to show one representation. Then ask what the grouping proves and what it does not prove.

|                |                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------|
| <b>Failure</b> | Outcome 2 - Messy Debate Without Anchor<br>Student voice is present, but the math has no anchor. |
|----------------|--------------------------------------------------------------------------------------------------|

- Typical path: Choice 1 = B/C + Choice 2 = A
- Key point: **Talk is not the same as reasoning.**
- Improve next: Pause the debate and introduce an anchor: a definition, a representation, or a counterexample test.

|                        |                                                                                                                               |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Partial Success</b> | <b>Outcome 3 - Definition Recovery</b><br>The teacher restores the definition, but the misconception is only partly resolved. |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------|

Typical path: Choice 1 = B/C + Choice 2 = B

Key point: Definition helps, but boundary testing is missing.

Improve next: After reviewing the definition, ask what would make a number odd and compare 5, 6, 9, and 10 on the board.

|                               |                                                                                                                       |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Strong Partial Success</b> | <b>Outcome 4 - Productive Contradiction</b><br>Sean’s idea becomes testable, but the class still needs consolidation. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------|

Typical path: Choice 1 = B/C + Choice 2 = C + Choice 3 = A/B

Key point: The misconception is treated as a hypothesis.

Improve next: After the contradiction, invite students to build and test the stronger leftover rule themselves.

|                |                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------|
| <b>Success</b> | <b>Outcome 5 - Conceptual Breakthrough</b><br>Students build and test a transferable odd/even rule. |
|----------------|-----------------------------------------------------------------------------------------------------|

Typical path: Choice 1 = B/C + Choice 2 = C + Choice 3 = C

Key point: The mistake becomes a resource for collective reasoning.

Improve next: Keep the final test student-owned: ask students to predict, represent, and explain the leftover rule.

9. Detailed result-page content by outcome

Use the following content for the debrief cards. Each section includes a highlighted key point so the result page is easier to scan.

Outcome 1 - Premature Closure

Performance level: Failure

General result summary: Correct, but closed. You protected the right answer, but the discussion ended before Sean’s reasoning became visible. The class may know that 6 is even, but they do not learn how to examine the mistaken rule behind Sean’s claim.

Sean: Sean’s thinking stays private. He hears that he is wrong, but he does not get help separating “6 can be made from groups” from “6 is classified as odd or even.”

Other students: The class loses a reasoning opportunity. Cassandra, Keith, Mei, and Ofala do not get to compare, test, or refine the idea.

Mathematical clarity: Accuracy is protected, but transfer is weak. Students receive the fact, not the method for deciding odd/even in future cases.

Improve next time: Ask for one representation before closing: “Sean, show us the three 2s. What does that show, and what does it not show?”

Outcome 2 - Messy Debate Without Anchor

Performance level: Failure

General result summary: Open discussion, weak structure. You made room for student voices, but the conversation became mostly classmates pushing back on Sean without a shared mathematical anchor.

Sean: Sean is challenged socially, not supported intellectually. He may defend his idea harder because the class is reacting to him instead of testing his reasoning.

Other students: Peer disagreement is useful but incomplete. Cassandra and Keith surface the problem, but Mei and Ofala need a clearer opening to move the room from arguing to reasoning.

Mathematical clarity: The concept stays loose. The class has energy, but not enough definition, representation, or counterexample work.

Improve next time: Before inviting more disagreement, say: "Pause. What definition are we using?" or "Let's test Sean's rule with another number."

### **Outcome 3 - Definition Recovery**

Performance level: Partial Success

General result summary: Stable, but not fully tested. You brought the class back to the definition of even numbers, which helps. The limitation is that Sean's alternate rule is not fully challenged.

Sean: Sean may still think both rules work. He can see that 6 fits the even definition, but he may still believe three groups of 2 makes it odd too.

Other students: The class gets a shared anchor. This prevents random debate, but students still need a chance to test and strengthen the rule themselves.

Mathematical clarity: Definition is necessary but not enough. Students need to see why group count is not the same as odd/even classification.

Improve next time: Add a contrast test: "What would make a number odd?" Then compare 5, 6, 9, and 10 using pairs and leftovers.

### **Outcome 4 - Productive Contradiction**

Performance level: Strong Partial Success

General result summary: Strong testing move. You treated Sean's reasoning as a hypothesis and tested it with a new case, so the class can see the contradiction.

Sean: Sean gets a reason to revise. He is not simply corrected; he sees that his rule would also make 10 both odd and even.

Other students: Mei's role becomes powerful. Her counterexample shifts the class from rejection to evidence.

Mathematical clarity: The misconception becomes visible. The board shows why a rule must work across cases, not only for 6.

Improve next time: After the counterexample, ask students to build the stronger rule: "So what does make a number odd?"

### **Outcome 5 - Conceptual Breakthrough**

Performance level: Success

General result summary: Best learning arc. You helped the class move from claim, to disagreement, to testing, to a stronger transferable rule.

Sean: Sean revises without losing face. His idea becomes the starting point for distinguishing group count from leftovers.

Other students: The class builds the rule together. Cassandra and Keith surface disagreement, Mei tests the logic, and Ofala offers the pairing-and-leftover rule.

Mathematical clarity: The rule transfers. Students learn that odd numbers leave 1 leftover after pairing by 2, while even numbers leave 0 leftover.

Improve next time: Even in this strong path, end with another student-generated number so the class owns the rule beyond this example.

## 10. Source note

This reference is based on the uploaded Sean/Ofala classroom transcript and the previous Sean's Number reference sheet. Key transcript moments used here include Sean's original claim about 6, peer disagreement, the class definition of even, Mei's test with 10, Ofala's pairing-and-leftover explanation, and the class test of the new rule.